(ST 623) Final Project - Generalized Linear Mixed Models

Introduction

Generalized Linear Mixed Models (GLMMs) are powerful statistical tools that extend both Linear Mixed Models (LMMs) and Generalized Linear Models (GLMs). GLMMs can find their place once either of the mentioned methods are not adequate.

LMMs rely on several key assumptions, including a linear relationship between the response and covariates, constant variance that is not a function of the mean, and Normally distributed random effects. When these assumptions are violated—such as when the variance depends on the mean or the data exhibit overdispersion—LMMs may fail to provide valid inferences.

Similarly, GLMs, while accommodating non-normal responses, cannot handle correlated or hierarchical data structures without additional modifications. GLMMs address these limitations by integrating random effects into the GLM framework, enabling the analysis of non-normal data with inherent dependencies or hierarchical structures. This makes GLMMs particularly effective for tackling issues like overdispersion, dependency violations, and other complexities encountered in experimental and observational studies.

This paper explores a brief look into the theoretical foundations then a deeper dive into simulation-based evaluations. The primary focus lies in understanding their flexibility, robustness, and practical utility across various scenarios via simulations.

The structure of this paper is as follows:

1. **Theory:** An overview of GLMMs, their components, and estimation techniques.
2. **Simulation Study:** An analysis using simulated data to explore the performance of GLMMs under different scenarios.
3. **Conclusion:** Summary of findings and insights gained from the project.

1 Theory

GLMMs extend the generalized linear model by incorporating random effects to account for hierarchical or correlated data structures. The general form of a GLMM is:

$$g(\mu) = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{Z}\boldsymbol{u},$$

where:

- $g(\mu)$ is the link function relating the mean response μ to the linear predictor,
- $\boldsymbol{X}$ is the design matrix for fixed effects, with the corresponding coefficients $\boldsymbol{\beta}^T = \begin{pmatrix}\beta_0 & \beta_1 & \cdots & \beta_{p-1}\end{pmatrix}$, where p is the number of fixed effects (including intercept).
- $\boldsymbol{Z}$ is the design matrix for random effects, and $\boldsymbol{u}$ represents the random effects coefficients. The random effects are defined by their intercepts and slopes, which can vary across the levels of factors.

Similar to fixed-effect GLMs, we can start with an exponential family of distributions to establish some nice relationships. However, now we now focus on the conditional probability density function (pdf), of our response given the random effects, $f_{Y_i|\boldsymbol{u}}(y_i|\boldsymbol{u})$, which results remain consistent with the GLM framework:

- **Conditional relationships:**

$$f_{Y_i|\boldsymbol{u}}(y_i|\boldsymbol{u}) = \exp\left\{\frac{y_i\theta_i - b(\theta_i)}{a_i(\phi)} - c(y_i, \phi)\right\}$$

$$E\left(Y_i|\boldsymbol{u}\right) = \frac{d}{d\theta}b(\theta_i) \equiv \mu_i \qquad \mathrm{Var}(Y_i|\boldsymbol{u}) = \frac{d^2}{d\theta_i^2}b(\theta_i)a_i(\phi) = V(\mu_i)a_i(\phi)$$

- **Unconditional relationships:** Through the use of Law of Total Expectation and Variance

$$E[Y_i] = E[E[Y_i|\boldsymbol{u}]] = E[\mu_i] = E\big[g^{-1}(\boldsymbol{X}\beta + \boldsymbol{Z}\boldsymbol{u})\big]$$

$$\mathrm{Var}(Y_i) = \mathrm{Var}\big(g^{-1}(\boldsymbol{X}\beta + \boldsymbol{Z}\boldsymbol{u})\big) + E\big[\mathrm{Var}(Y_i|\boldsymbol{u})\big]$$

2 Simulation Study

2.1 Overview of the Data Generation Function

The data generation function, is designed to simulate data for generalized linear mixed models (GLMMs) with both fixed and random effects. It provides flexibility in specifying various aspects of the model, including the number of fixed and random effects, the distribution of the response variable, the correlation structure of the covariates, and overdispersion inducing options are available.

We will begin with a 'base case' dataset, from which we will incrementally modify specific parameters. This approach isolates and clarifies the individual effects of each change on the results. The simulation consists of 300 runs with a sample size of $N = 240$. The fixed effects coefficients are $\boldsymbol{\beta} = [0.25, 0.50, 0.75]$, corresponding to the intercept and two covariates. Covariates X_1 and X_2 are generated from a Multivariate Normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$

$$\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \boldsymbol{\Sigma} = \begin{pmatrix} 1.00 & 0.50 \\ 0.50 & 1.00 \end{pmatrix}$$

For random effects, two variables (Z_1, Z_2) are included, with 4 and 5 levels, respectively. The common standard deviations for random intercepts and slopes are $\sigma_{\text{int}} = [0.50, 0.50]$ and $\sigma_{\text{slope}} = [1, 1]$. The response variable follows a Poisson distribution with a log link function:

$$\text{Response} = Y \sim \text{Poisson}(\lambda), \quad \lambda = \exp(\eta),$$

where η is the linear predictor. No noise or overdispersion is introduced yet. Using similar notation as the previous section, the linear predictor is defined as:

$$\eta_i = \sum_{j=1}^{k} \beta_j X_{ij} + \sum_{m=1}^{q} Z_{im} u_m,$$

The simulation study aims to explore the performance of GLMMs under varying conditions. Key aspects include:

(1) Examining GLMM performance across different data generation settings, starting with the base case and analyzing the effects of:

- Increasing sample size
- Adding additional levels to the random effects

(2) Investigating GLM fits on the same data:

- Evaluating the impact of excluding true random effect estimates
- Investigating model fits where random effects are treated as fixed factors
  - Assessing the consequences of not fitting a saturated model in this context

(3) Assessing the robustness of GLMMs:

- Initially intended to examine overdispersion but evolved into an evaluation of GLMM robustness
- Testing GLMM robustness by adding varying levels of noise to the data

Before each run of simulations, a common seed was set to ensure reproducibility and facilitate meaningful comparisons. This approach allows us to simulate alternative scenarios, akin to exploring parallel universes. By maintaining consistent initial conditions, we can evaluate how specific actions or changes lead to the observed results, making the comparisons more reliable and insightful.

2.2 Results

(1) The results from the base case scenario simulation are presented in Table 1, Figure 1, and Figure 2. We observe that while the fixed effect estimates are reasonably accurate and unbiased, the random effect coefficients deviate by approximately 12 to 18% from the true values, this discrepancy is likely not entirely due to simulation error. Lastly we see the average overdispersion estimate is approximately 0.90, close to the expected value of one.

To further investigate, we increased the sample size by a factor of 10. The results from these simulations are presented in Table 2, Figure 3, and Figure 4. Interestingly, this substantial increase in sample size does not significantly impact the standard errors of the fixed effects or the estimates of the random effects. Any observed differences are minimal and could be attributed to simulation error, although there might be a slight reduction in the fixed effects' standard errors. Overall, Tables 1 and 2 show comparable results, with no substantial differences observed.

Reverting the sample size to the base case of 240, we increase the number of levels in both random effects: the first from 4 to 40 levels and the second from 5 to 60 levels. The results are summarized in Table 3, Figure 5, and Figure 6. The most notable change is a significant reduction in the fixed effect standard errors, which are now more than halved. Additionally, the random effect estimates are much closer to the true values. The average overdispersion estimate has also decreased, approximately halving to 0.44.

With the higher random effect level counts maintained, increasing the sample size again to 2400 produces the ideal scenario for GLMMs: all estimates are very close to the true values, and standard errors are relatively low. The average overdispersion estimate increases to approximately 0.82. These results are detailed in Table 4, Figure 7, and Figure 8.

(2) For the following GLM simulations, 50,000 runs were performed due to the significantly lower computational time required compared to GLMM fittings. The results for excluding the random effects from the model are presented in Table 5 and Figure 9. This exclusion results in highly inaccurate estimates and unrealistically low standard errors. Compared to the base case scenario, the standard errors are approximately 36 times smaller, leading to extremely large test statistics. Additionally, the average overdispersion parameter is around 49,660, which is far greater than one and indicates severe model misfit. These issues become even more pronounced as the number of levels for each random effect increases.

When fitting the GLM with the random effect variables treated as fixed effect factors and including all interactions, the results are shown in Table 6, Figure 10, Figure 11, and Figure 12. In this approach, we obtain unbiased estimates for the fixed effects $\hat{\beta}_1$ and $\hat{\beta}_2$; however, the standard errors are still approximately five times smaller than those from the GLMM base case fit. Increasing the sample size to 2,400 further reduces the standard errors, as expected, as shown in Table 7. For both scenarios described, all factor effects and interaction estimates are centered around and close to 0. Furthermore, the average overdispersion parameter in these models aligns closely with their GLMM counterparts.

When the saturated model is not specified, we observe biased estimates for $\hat{\beta}_1$ and $\hat{\beta}_2$, as shown in Table 8 and Figure 13. Additionally, the standard errors are severely underestimated, resulting in excessively large z-values. The average overdispersion parameter in this case is approximately 78, indicating significant model misfit.

Increasing the number of levels for each random effect leads to slight improvements, but not to an acceptable degree, as demonstrated in Table 9 and Figure 14. Although the random effect estimates are fit, they are omitted here for brevity; all estimates are close to 0 with high standard errors ranging from approximately 6 to 40. The overdispersion parameter decreases but remains high at around 12.15.

When fitting the base case model while treating the random effects as continuous fixed effects, we encounter extreme results, as shown in Table 10. Specifically, the average z-values are orders of magnitude higher than expected, and the average overdispersion estimate is an exceptionally large value of 1.746332e+45. Increasing the number of levels further exacerbates these extreme values. As before, the fixed effect estimates remain biased.

Throughout the fitting process, numerous issues arise, including problems with convergence and instances where fitted rates are numerically zero. Additionally, when random effects are treated as continuous fixed effects, occasional NA values are encountered due to singularities.

(3) An initial attempt to induce overdispersion using various hierarchical structures is detailed in Tables 11, 12, and 13. Three structures were explored with arbitrary parameter settings: Poisson-Poisson, Gamma-Poisson, and Normal-Poisson. However, the results countered expectations, with the overdispersion parameter consistently estimated at approximately 0.45 across all structures. While some biased estimates of effects were observed—least notably in the Gamma-Poisson case—the biases were not particularly severe. The only noteworthy issue was occasional convergence problems with the fitting algorithm.

Lastly, we evaluate the robustness of GLMMs under a "best-case scenario" by introducing varying levels of noise. Starting with $\epsilon_i \sim N(0, \sigma^2 = 4)$, we progressively increase σ^2 to 25 while simultaneously scaling the sample size by a factor of 10. These modifications build upon the base case, incorporating an increased number of levels for the random effects.

The results are presented in Tables 14 and Figures 15 and 16. Introducing this small amount of noise results in substantial biases across all estimated coefficients while reducing the standard errors by approximately 25% compared to the results shown previously in Table 3. The figures highlight the shifted sampling distribution of the estimated coefficients from their true values. When both the sample size and error variance are increased simultaneously, the results remain remarkably consistent. While figures for these results are omitted due to their extreme similarity, they are summarized in Table 15. A difference unseen by either the tables or figures is the overdispersion estimates which actually increased from approximately 0.50 to 3.5 when increasing the sample size and error in sync.

2.3 Discussion

(1) The simulations demonstrate that for GLMMs to yield accurate coefficient estimates, a sufficient number of levels within the random effects is crucial. Merely increasing the sample size does not meaningfully enhance precision compared to increasing the number of levels. This underscores the importance of appropriately capturing the hierarchical structure of the data when specifying the model.

(2) Predictably, failing to account for significant random effects leads to inflated overdispersion parameters and severely underestimated standard errors, which can create a false sense of confidence in statistical inference. Modeling random effects as fixed factors appears to perform well as an alternative, provided a saturated model is specified. Excluding interactions or simplifying the model introduces bias into the results.

    Catastrophic issues arise when treating random effects as fixed continuous variables. This approach produces biased estimates, computational challenges, and convergence problems, further highlighting the limitations of such a method.

    One of the most intriguing findings is the comparability between GLMMs and saturated GLMs that treat random effects as fixed factors. Saturated GLMs produce much smaller standard errors, raising questions about the practical advantage of GLMMs beyond conserving degrees of freedom. This observation suggests a potential avenue for further research to better understand the trade-offs between these modeling approaches.

(3) Although the attempt to induce overdispersion did not yield the expected results, it highlighted the semi-robust nature of GLMMs to changes in the response variable. Conversely, introducing variability into the response through different levels of noise revealed substantial biases in the estimates, underscoring the sensitivity of GLMMs to noise in the data.

    These findings emphasize the need for careful consideration when fitting GLMMs, particularly in the presence of unaccounted variability or poorly specified random effects.

3 Conclusion

This project examined the flexibility and robustness of GLMMs by exploring their response to various data settings. Through simulation studies, we adjusted key parameters—such as sample size, random effect levels, and noise—to assess their impact on GLMM performance. These explorations revealed scenarios where GLMMs excel and instances where their use is less appropriate, ultimately contributing to a deeper understanding and more effective application of these complex models.

4 Appendix

4.1 Tables

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	**z-value**
(Intercept)	0.2479202	0.2989986	0.9560278
X_1	0.4701891	0.5876457	0.8830049
X_2	0.7550062	0.5835128	1.457646

Random Effects	
Grouping	**Std.Dev.**
Z_1 **(Intercept)**	0.4428584
Z_1 (X_1)	0.8788932
Z_1 (X_2)	0.8580535
Z_2 **(Intercept)**	0.3987277
Z_2 (X_1)	0.8212244
Z_2 (X_2)	0.8252235

Table 1: Model Summaries (Base Case): Fixed and Random Effects Estimates Averaged Across 300 Simulations

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	**z-value**
(Intercept)	0.2414686	0.2912030	0.9055616
X_1	0.4760420	0.5624542	0.9804580
X_2	0.7699397	0.5735347	1.5790281

Random Effects	
Grouping	**Std.Dev.**
Z_1 **(Intercept)**	0.4473965
Z_1 (X_1)	0.8564518
Z_1 (X_2)	0.9067077
Z_2 **(Intercept)**	0.4072396
Z_2 (X_1)	0.8447616
Z_2 (X_2)	0.8326552

Table 2: Model Summaries (Increased Sample Size from 240 to 2400): Fixed and Random Effects Estimates Averaged Across 300 Simulations

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	**z-value**
(Intercept)	0.2481687	0.1430899	1.815502
X_1	0.4972333	0.2398306	2.090667
X_2	0.7711701	0.2421858	3.200430

Random Effects	
Grouping	**Std.Dev.**
Z_1 **(Intercept)**	0.4650269
Z_1 (X_1)	0.9867416
Z_1 (X_2)	0.9677326
Z_2 **(Intercept)**	0.4157126
Z_2 (X_1)	0.9816241
Z_2 (X_2)	1.0054306

Table 3: Model Summaries (Sample Size 240, Random Effect Level Counts Increase $4 \to 40, 5 \to 60$): Fixed and Random Effects Estimates Averaged Across 300 Simulations

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	**z-value**
(Intercept)	0.2547493	0.1026946	2.497555
X_1	0.4932621	0.2031357	2.442682
X_2	0.7404413	0.2036049	3.662236

Random Effects	
Grouping	**Std.Dev.**
Z_1 **(Intercept)**	0.4936928
Z_1 (X_1)	0.9896218
Z_1 (X_2)	1.0055527
Z_2 **(Intercept)**	0.4925471
Z_2 (X_1)	0.9952355
Z_2 (X_2)	0.9883506

Table 4: Model Summaries (Sample Size 2400, Random Effect Level Counts $40, 60$): Fixed and Random Effects Estimates Averaged Across 300 Simulations

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	z**-value**
(Intercept)	0.8413958	0.03093496	62.22856
X_1	0.7591496	0.01660387	86.94507
X_2	1.1538891	0.01691860	129.51984

Table 5: Model Summaries (Exclusion of Random Effects): Fixed Effects Averaged Across 50,000 Simulations

	(Intercept)	X_1	X_2	$(Z_1)2$	$(Z_1)3$	$(Z_1)4$	$(Z_2)2$	$(Z_2)3$
Estimate	0.2314	0.5010	0.7465	0.0033	-0.0005	-0.0010	-0.0083	-0.0042
Std.Error	0.1527	0.1095	0.1110	0.1478	0.1480	0.1480	0.1696	0.1695
z**-value**	2.8756	6.2334	9.2115	0.0226	-0.0034	-0.0063	-0.0546	-0.0259

	$(Z_2)4$	$(Z_2)5$	$(X_1{:}Z_1)2$	$(X_1{:}Z_1)3$	$(X_1{:}Z_1)4$	$(X_2{:}Z_1)2$	$(X_2{:}Z_1)3$	$(X_2{:}Z_1)4$
Estimate	-0.0045	-0.0001	-0.0014	-0.0004	0.0015	-0.0017	0.0055	0.0095
Std.Error	0.1694	0.1693	0.1072	0.1072	0.1072	0.1088	0.1088	0.1090
z**-value**	-0.0356	-0.0057	0.0160	-0.0060	0.0179	-0.0235	0.0689	0.0987

	$(X_1{:}Z_2)2$	$(X_1{:}Z_2)3$	$(X_1{:}Z_2)4$	$(X_1{:}Z_2)5$	$(X_2{:}Z_2)2$	$(X_2{:}Z_2)3$	$(X_2{:}Z_2)4$	$(X_2{:}Z_2)5$
Estimate	-0.0029	0.0040	-0.0005	-0.0084	0.0037	0.0033	0.0055	0.0019
Std.Error	0.1235	0.1234	0.1232	0.1232	0.1251	0.1251	0.1249	0.1249
z**-value**	-0.0240	0.0542	-0.0058	-0.0508	0.0317	0.0272	0.0497	-0.0367

Table 6: Model Summaries (Saturated Model - Fitting Random Effects as Fixed Factors): Fixed Effects Averaged Across 50,000 Simulations

	(Intercept)	X_1	X_2	$(Z_1)2$	$(Z_1)3$	$(Z_1)4$	$(Z_2)2$	$(Z_2)3$
Estimate	0.249964	0.510980	0.754204	-0.003298	-0.001241	-0.001790	0.002839	-0.000393
Std.Error	0.037305	0.020447	0.020665	0.036140	0.036081	0.036093	0.041506	0.041540
z**-value**	12.915138	51.048815	72.636868	-0.087801	-0.069944	-0.057963	0.120346	0.012719

	$(Z_2)4$	$(Z_2)5$	$(X_1{:}Z_1)2$	$(X_1{:}Z_1)3$	$(X_1{:}Z_1)4$	$(X_2{:}Z_1)2$	$(X_2{:}Z_1)3$	$(X_2{:}Z_1)4$
Estimate	-0.002249	-0.000701	-0.007399	0.000852	-0.002105	-0.019880	-0.007373	-0.004590
Std.Error	0.041530	0.041524	0.019694	0.019647	0.019649	0.019946	0.019882	0.019902
z**-value**	-0.073641	-0.043671	0.135523	0.185963	0.408815	-0.787718	-0.288077	-0.334913

	$(X_1{:}Z_2)2$	$(X_1{:}Z_2)3$	$(X_1{:}Z_2)4$	$(X_1{:}Z_2)5$	$(X_2{:}Z_2)2$	$(X_2{:}Z_2)3$	$(X_2{:}Z_2)4$	$(X_2{:}Z_2)5$
Estimate	-0.005249	-0.004454	0.003343	-0.009200	0.001033	-0.000609	-0.002657	-0.003141
Std.Error	0.022890	0.022926	0.022928	0.022908	0.023134	0.023175	0.023181	0.023157
z**-value**	-0.291081	-0.175260	0.220446	-0.501061	-0.035138	-0.318115	-0.030624	-0.391900

Table 7: Model Summaries (Saturated Model - Fitting Random Effects as Fixed Factors - Increased Sample Size 240 to 2400): Fixed Effects Averaged Across 50,000 Simulations

	(Intercept)	X_1	X_2
Estimate	0.09499857	0.73352749	1.11661578
Std.Error	0.08079868	0.02022436	0.02079535
z-value	12.35533695	47.84187779	70.32569453

	$(Z_1)2$	$(Z_1)3$	$(Z_1)4$	$(Z_2)2$	$(Z_2)3$	$(Z_2)4$	$(Z_2)5$
Estimate	0.007411458	0.01387237	0.00575769	0.004671795	0.001589022	0.004607725	0.003040873
Std.Error	0.071746451	0.07159654	0.07173369	0.083305462	0.083208193	0.083120827	0.083094865
z-value	-0.030466874	0.17628056	0.02026673	0.047444162	0.104786527	0.038663737	-0.098490220

Table 8: Model Summaries (Model without Interactions - Fitting Random Effects as Fixed Factors): Fixed Effects Averaged Across 50,000 Simulations

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	**z-value**
(Intercept)	-0.3435612	23.6713466	1.5166433
X_1	0.63765452	0.05279072	12.03349401
X_2	0.95313711	0.05435761	17.54690093

Table 9: Model Summaries (Model without Interactions (Increased levels) - Fitting Random Effects as Fixed Factors): Fixed Effects Averaged Across 50,000 Simulations

	(Intercept)	X_1	X_2
Estimate	0.1649745	0.72143830	1.0916598
Std.Error	0.1193706	0.07149645	0.0728833
z-value	2.166288e+12	-4.715251e+11	-2.663817e+12

	Z_1	Z_2	X_1:Z_1	X_1:Z_2	X_2:Z_1	X_2:Z_2
Estimate	0.008962531	0.01046508	-0.002179156	-0.00362039	-0.002049104	-0.003002394
Std.Error	0.032820678	0.02544473	0.019805466	0.01547523	0.020239873	0.015783915
z-value	-4.844742e+11	-3.651257e+12	5.198015e+10	5.477968e+12	6.286589e+11	2.878267e+12

Table 10: Model Summaries - Fitting Random Effects as Fixed Continuous Variables with Interactions): Fixed Effects Averaged Across 50,000 Simulations

- Poisson-Poisson
  - $W_i = \sum_{j=1}^{y_i} Y_{ij}$, where $Y_{ij} \sim \text{Poisson}(\lambda)$, a choice of $\lambda = 3$ was arbitrary chosen. $y_i \sim \text{Poisson}(\eta)$ where η represents the linear predictors, as before

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	z**-value**
(Intercept)	0.8988493	0.2035819	18.575540
X_1	0.6145881	0.3007008	8.106268
X_2	0.9203344	0.3023670	9.826325

Random Effects	
Grouping	**Std.Dev.**
Z_1 **(Intercept)**	0.8487832
Z_1 (X_1)	1.3378340
Z_1 (X_2)	1.3362689
Z_2 **(Intercept)**	0.7811653
Z_2 (X_1)	1.2680037
Z_2 (X_2)	1.2799103

Table 11: Model Summaries (Poisson-Poisson Hierarchy): Fixed and Random Effects Averaged Across 300 Simulations

- Gamma-Poisson
  - $W_i = \sum_{j=1}^{y_i} Y_{ij}$, where $Y_{ij} \sim \text{Gamma}(\text{shape} = k, \text{rate} = \theta)$, values of k, θ were arbitrarily chosen to be 5 and 3 respectively and as before $y_i \sim \text{Poisson}(\eta)$

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	z**-value**
(Intercept)	0.5914585	0.1662281	6.405307
X_1	0.5230467	0.2604506	2.872826
X_2	0.8077014	0.2618145	6.159927

Random Effects	
Grouping	**Std.Dev.**
Z_1 **(Intercept)**	0.6385642
Z_1 (X_1)	1.1062898
Z_1 (X_2)	1.1220436
Z_2 **(Intercept)**	0.5775024
Z_2 (X_1)	1.0764564
Z_2 (X_2)	1.0690188

Table 12: Model Summaries (Gamma-Poisson Hierarchy): Fixed and Random Effects Averaged Across 300 Simulations

- Normal-Poisson
  - $W_i = \sum_{j=1}^{y_i} Y_{ij}$, where $Y_{ij} \sim \mathcal{N}(\mu, \sigma^2)$, values of μ, σ were arbitrarily chosen to be 6 and 2 respectively and as before $y_i \sim \text{Poisson}(\eta)$

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	z**-value**
(Intercept)	1.3550063	0.2262129	121.08848
X_1	0.6997335	0.3281640	47.15083
X_2	1.0460726	0.3314858	73.95544

Random Effects	
Grouping	**Std.Dev.**
Z_1 **(Intercept)**	1.057820
Z_1 (X_1)	1.591120
Z_1 (X_2)	1.605579
Z_2 **(Intercept)**	0.9514661
Z_2 (X_1)	1.4810924
Z_2 (X_2)	1.4958453

Table 13: Model Summaries (Normal-Poisson Hierarchy): Fixed and Random Effects Averaged Across 300 Simulations

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	z**-value**
(Intercept)	1.2567984	0.08246219	84.12694
X_1	0.2677178	0.14101835	14.41189
X_2	0.3930253	0.14327640	22.25987

Random Effects	
Grouping	**Std.Dev.**
Z_1 **(Intercept)**	0.3202029
Z_1 (X_1)	0.6577295
Z_1 (X_2)	0.6803078
Z_2 **(Intercept)**	0.3031674
Z_2 (X_1)	0.6191926
Z_2 (X_2)	0.6257730

Table 14: Model Summaries $\epsilon_i \sim N(0, 4)$: Fixed and Random Effects Averaged Across 300 Simulations

Fixed Effects			
Parameter	**Estimate**	**Std.Error**	z**-value**
(Intercept)	1.5532722	0.05662005	27.682818
X_1	0.3027745	0.14650621	2.065643
X_2	0.4708479	0.14860221	3.169877

Random Effects	
Grouping	**Std.Dev.**
Z_1 **(Intercept)**	0.2801399
Z_1 (X_1)	0.7195916
Z_1 (X_2)	0.7360944
Z_2 **(Intercept)**	0.2672423
Z_2 (X_1)	0.7120920
Z_2 (X_2)	0.7184686

Table 15: Model Summaries $\epsilon_i \sim N(0, 25)$ and Sample Size 2400: Fixed and Random Effects Averaged Across 300 Simulations

4.2 Figures

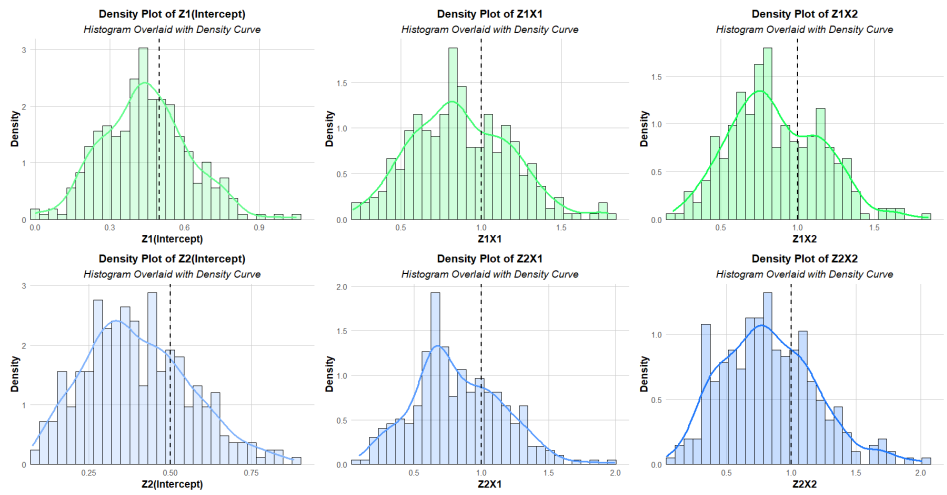

Figure 1: (Base Case): Random Effect Estimate Sampling Distributions

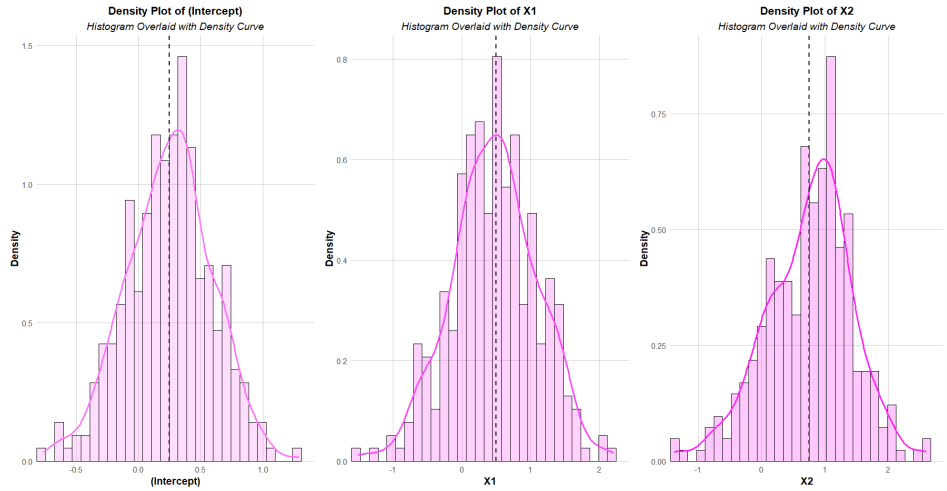

Figure 2: (Base Case): Fixed Effect Estimate Sampling Distributions

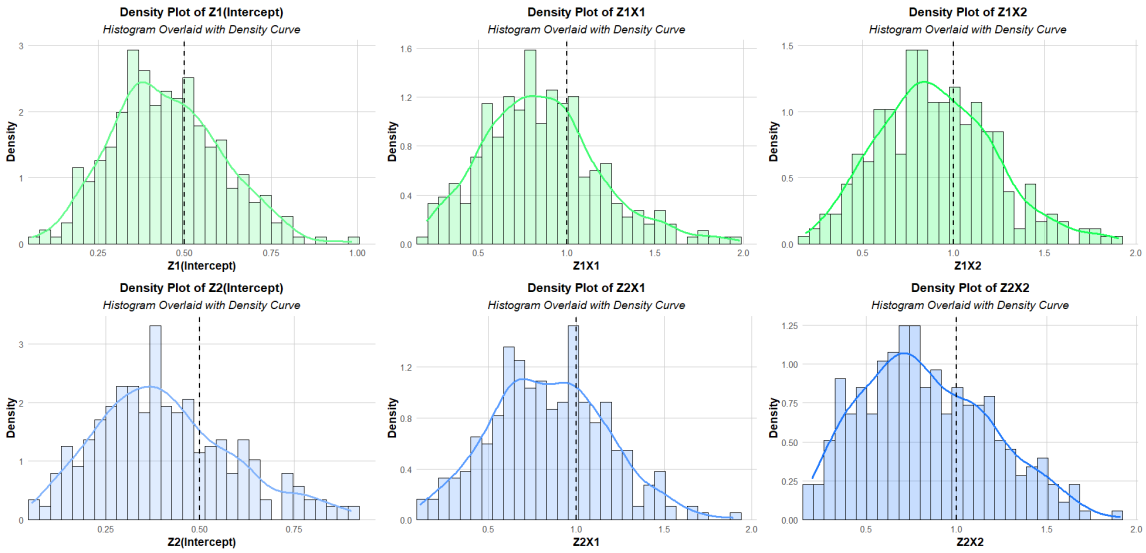

Figure 3: (Increased Sample Size from 240 to 2400): Random Effect Estimate Sampling Distributions

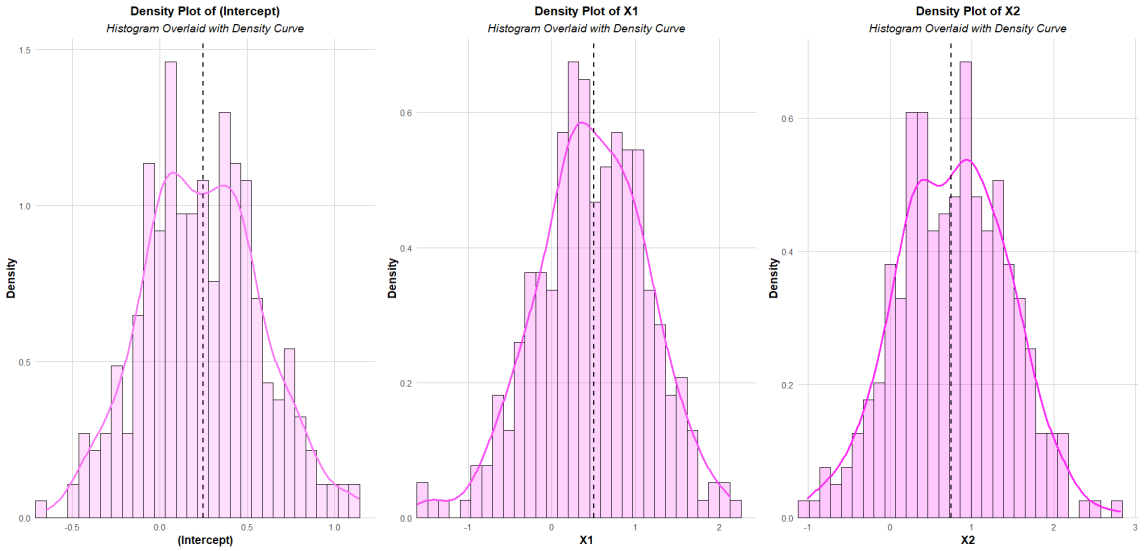

Figure 4: (Increased Sample Size from 240 to 2400): Fixed Effect Estimate Sampling Distributions

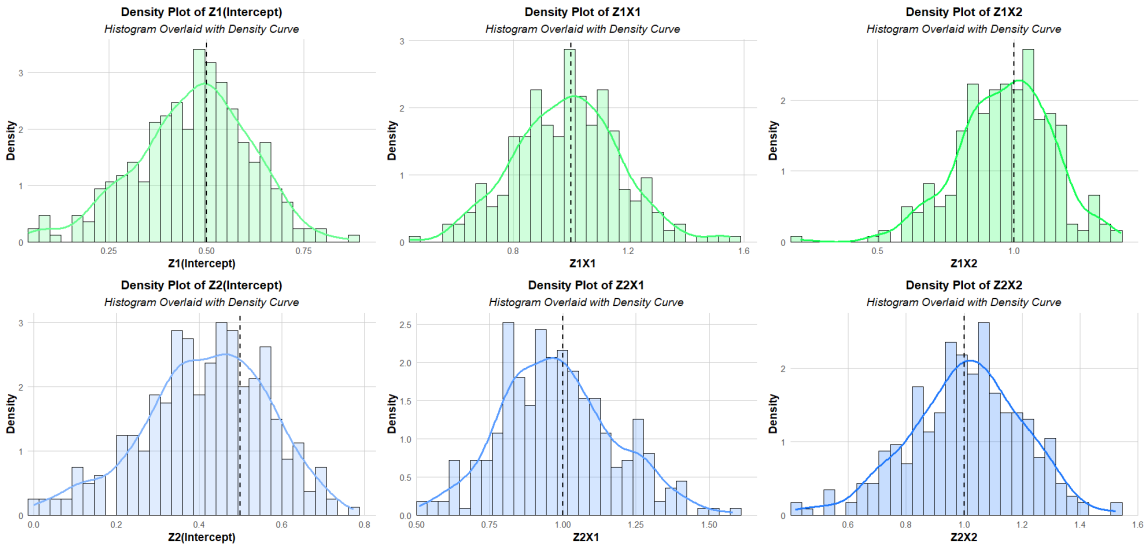

Figure 5: (Sample Size 240, Random Effect Level Counts Increase $4 \rightarrow 40, 5 \rightarrow 60$): Random Effect Estimate Sampling Distributions

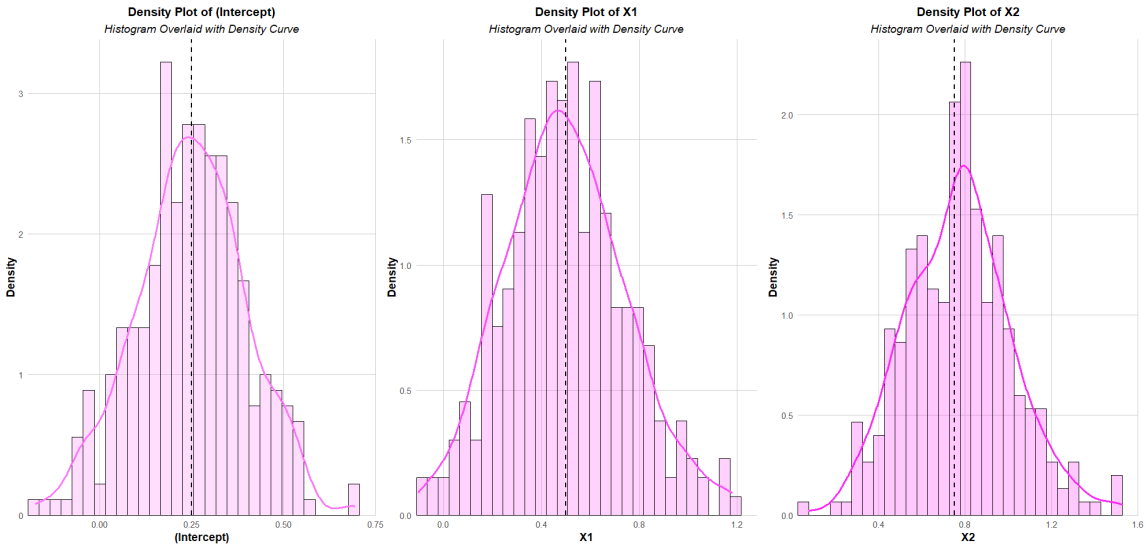

Figure 6: (Sample Size 240, Random Effect Level Counts Increase $4 \rightarrow 40, 5 \rightarrow 60$): Fixed Effect Estimate Sampling Distributions

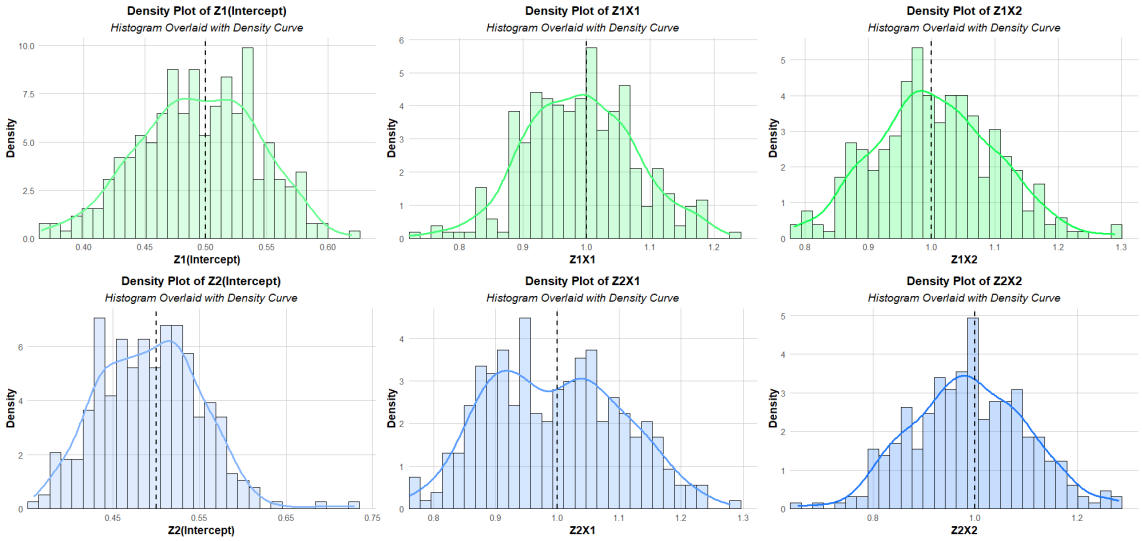

Figure 7: (Sample Size 2400, Random Effect Level Counts $40, 60$): Random Effect Estimate Sampling Distributions

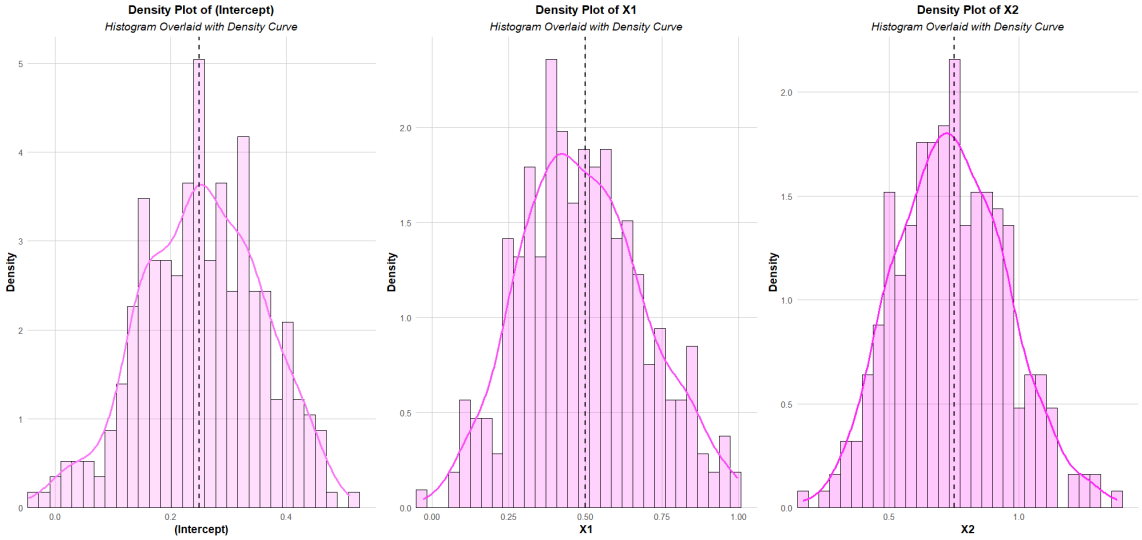

Figure 8: (Sample Size 2400, Random Effect Level Counts $40, 60$): Fixed Effect Estimate Sampling Distributions

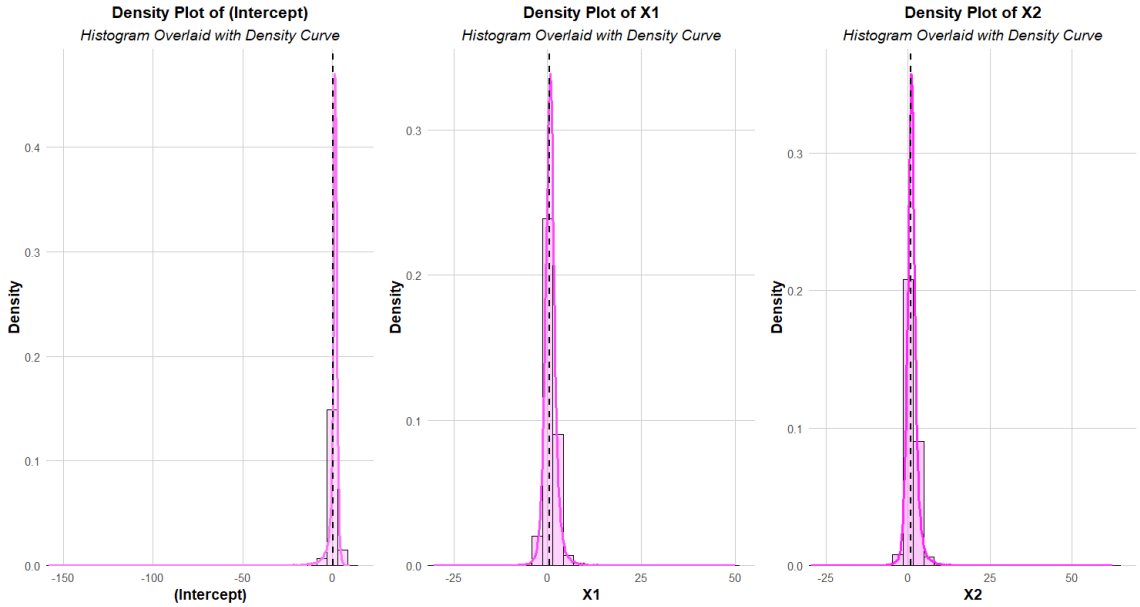

Figure 9: (Exclusion of True Random Effects): Fixed Effect Estimate Sampling Distributions

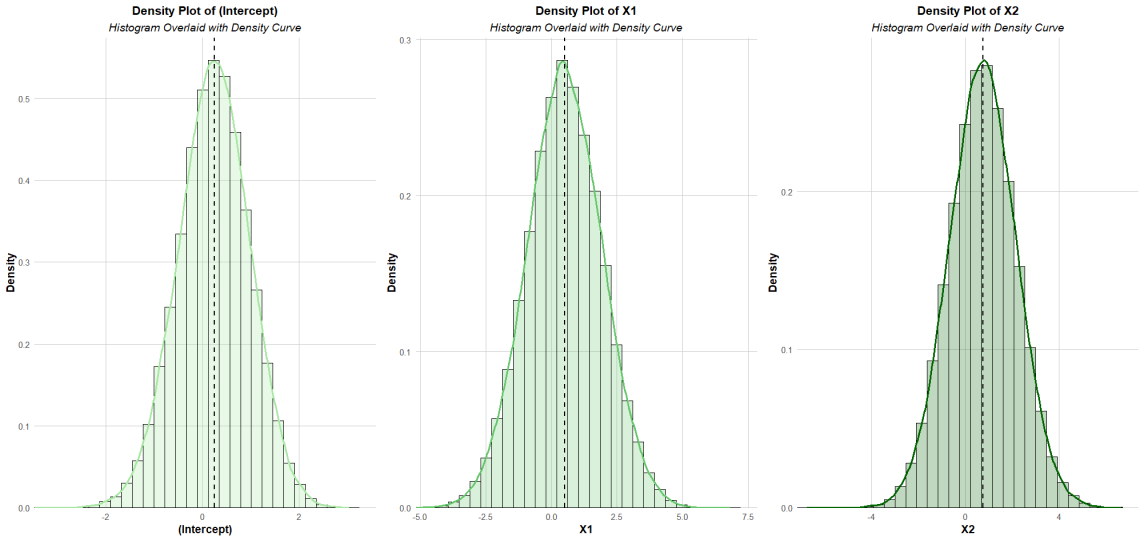

Figure 10: (Saturated Model - Fitting Random Effects as Fixed Factors): Fixed Effect Estimate Sampling Distributions

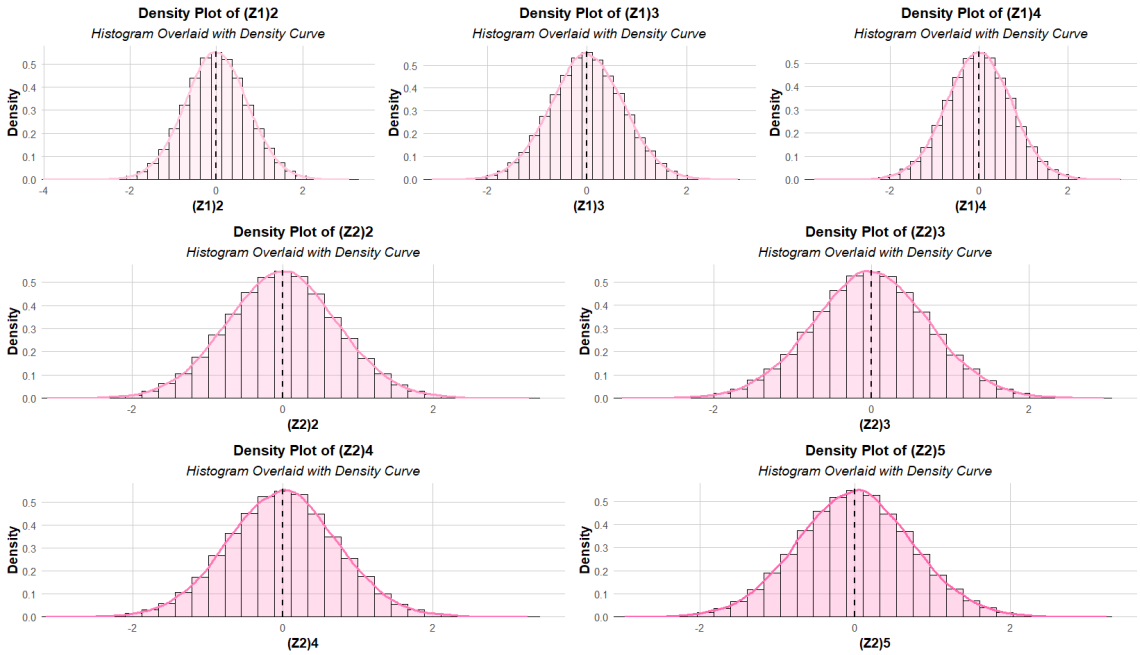

Figure 11: (Saturated Model - Fitting Random Effects as Fixed Factors): Fixed Effect Estimate Sampling Distributions for Factors

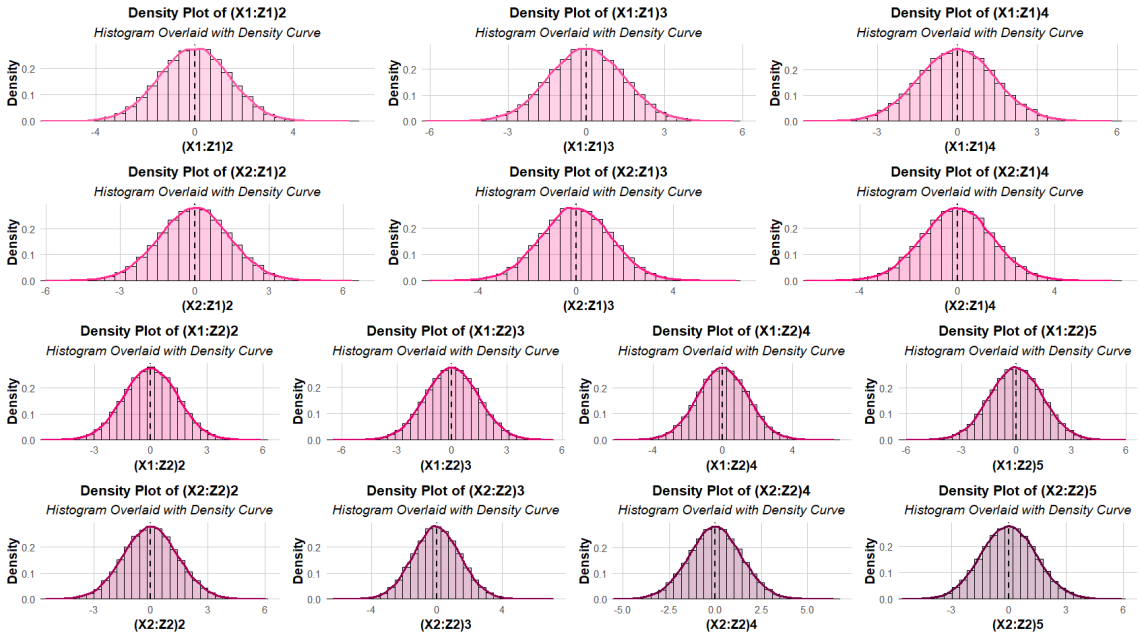

Figure 12: (Saturated Model - Fitting Random Effects as Fixed Factors): Fixed Effect Estimate Sampling Distributions for all Interactions

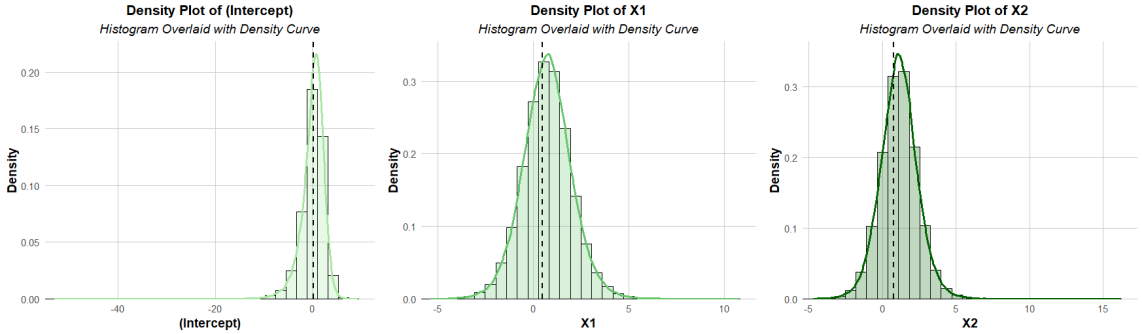

Figure 13: (Fitting Random Effects as Fixed Factors Excluding Interactions): Fixed Effect Estimate Sampling Distributions for Factors

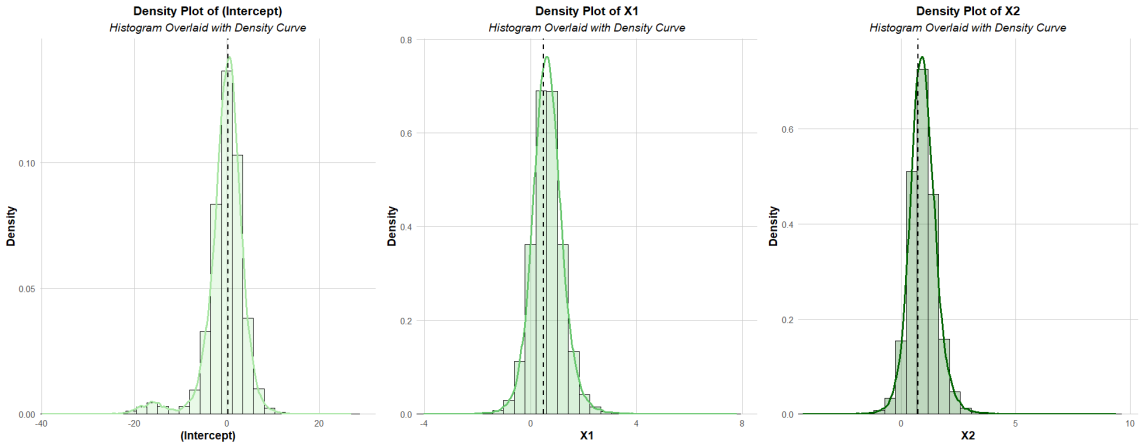

Figure 14: (Fitting Random Effects as Fixed Factors Excluding Interactions, More levels): Fixed Effect Estimate Sampling Distributions for all Interactions

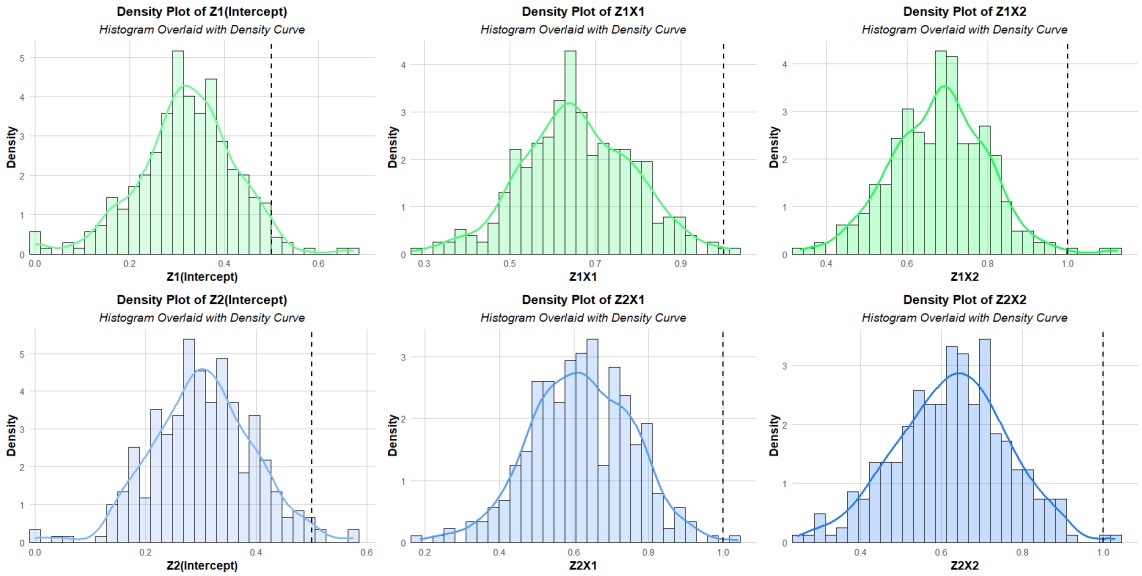

Figure 15: $\epsilon_i \sim N(0, 4)$: Random Effect Estimate Sampling Distributions

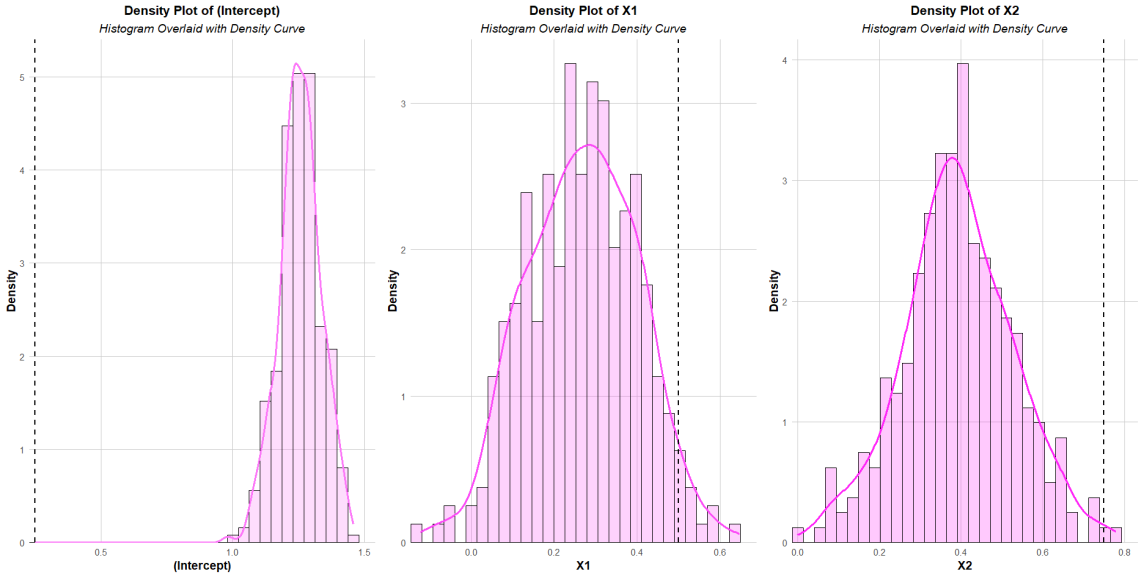

Figure 16: $\epsilon_i \sim N(0, 4)$: Fixed Effect Estimate Sampling Distributions